

SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Concept Mapping

COURSE NAME: Data Structures

COURSE CODE: CSA0309

Register Number	192572012
Name	Deepthi Korrapati

COURSE OUTCOME COVERED

CO1: Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL2, BL3)

Assessment Tool Weightage: 35%

Questions:2 EACH QUESTIONS CARRIES: 10 MARKS Total Marks:20 Duration: ONE HOUR

Q1 A smart healthcare system processes large volumes of patient data such as temperature readings, medical history, and diagnostic details. Construct a concept map linking the progression from Data → Data Object → Data Structure → Abstract Data Type (ADT) → Implementation in C. Identify relevant sub-concepts such as structures, pointers, and operations, and establish meaningful relationships among them. Organize the concept map hierarchically and justify how each level contributes to efficient data organization, processing, and improved system performance.

Q2 A smart city traffic management system handles data related to vehicle movement, road networks, and signal control. Construct a concept map connecting Primitive Data Structures → Non-Primitive Data Structures → Linear → Non-Linear Structures. Include examples such as arrays, linked lists, trees, and graphs, and illustrate their applications in traffic management. Explain the relationships between these structures and justify their classification based on functionality and efficiency.

Code of Conduct:

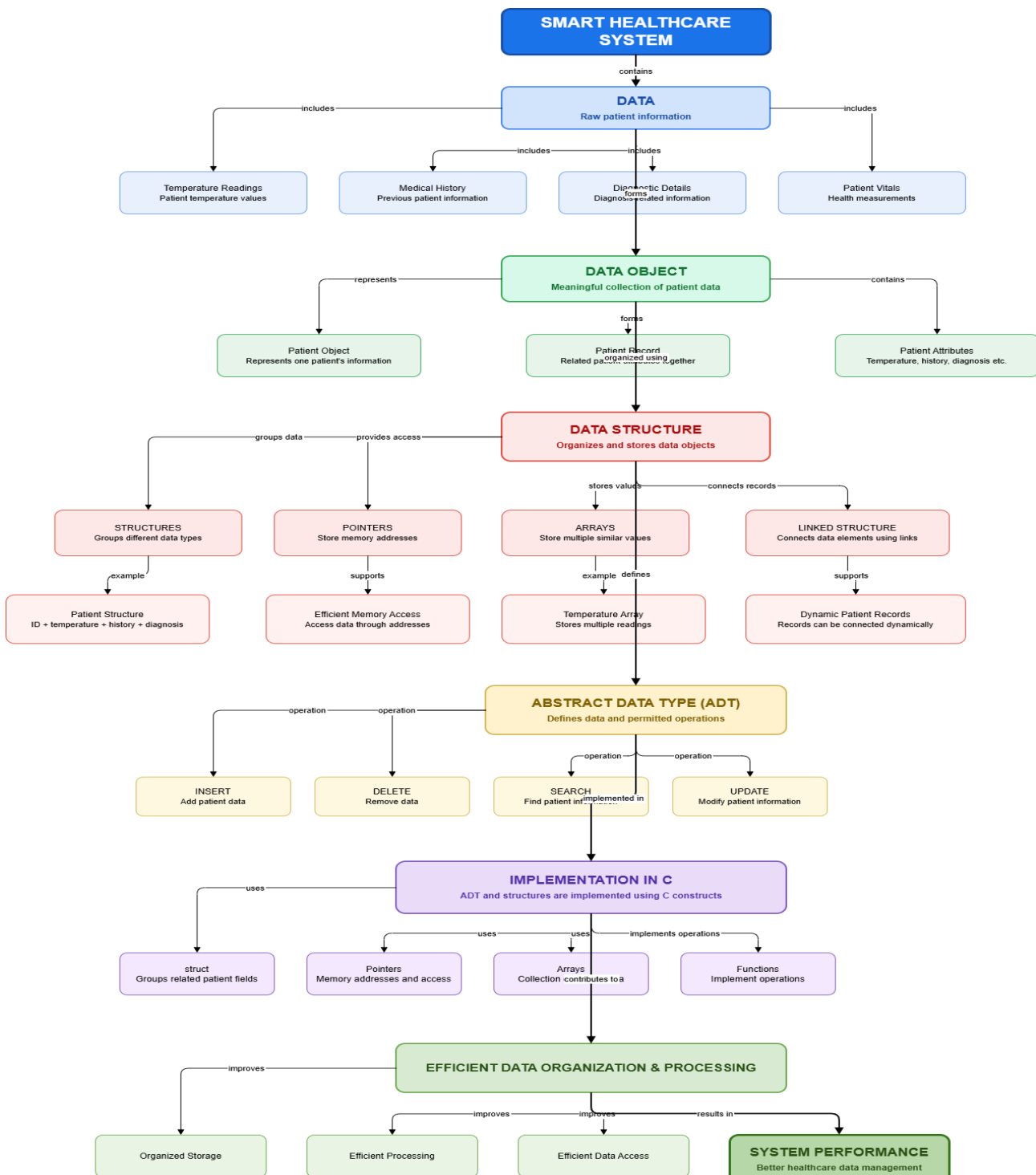
I DEEPTHI KORRAPATI(192572012) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Deepthi

RUBRICS FOR CALCULATION:

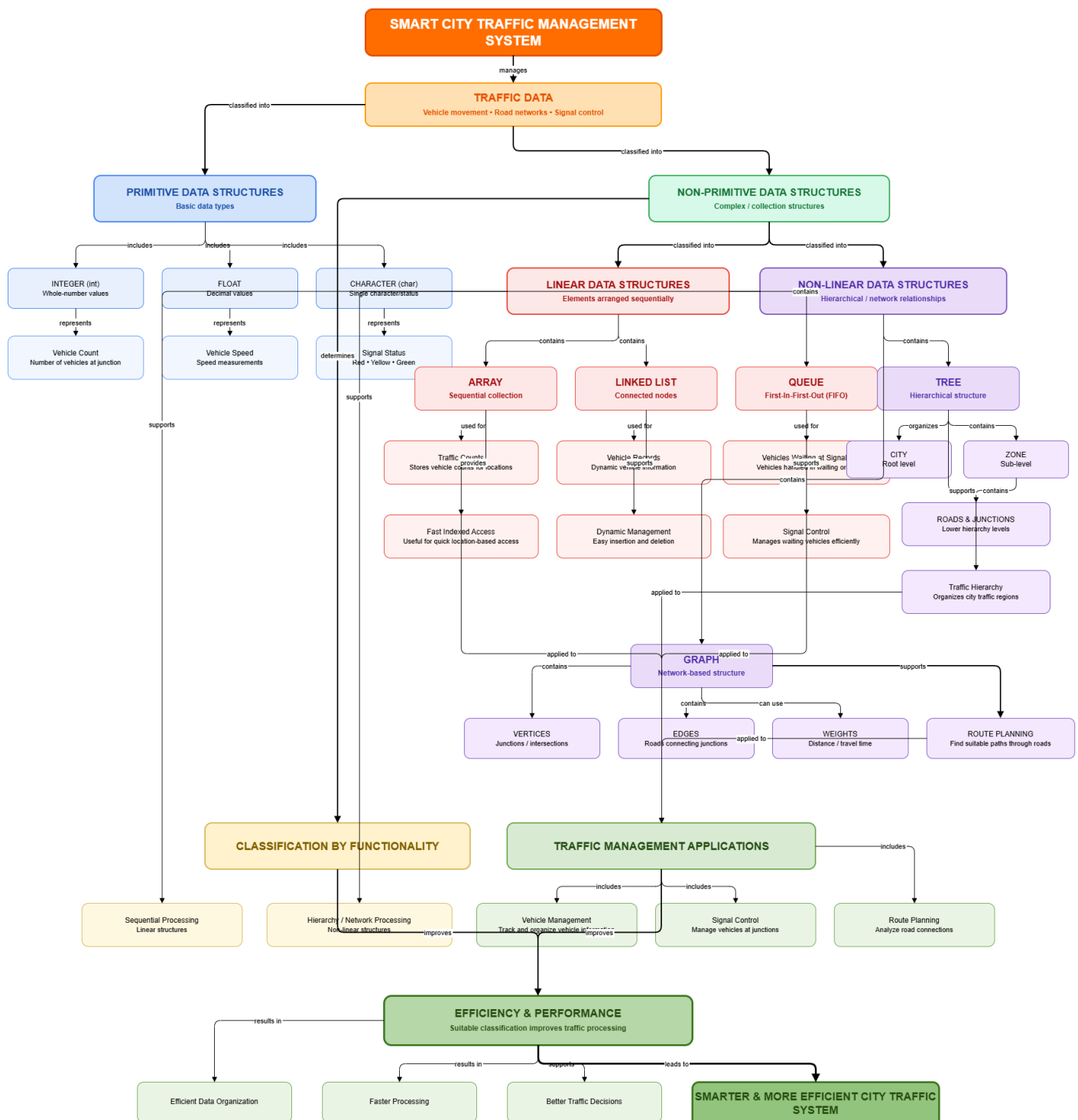
Concept Mapping					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Q1 A smart healthcare system processes large volumes of patient data such as temperature readings, medical history, and diagnostic details. Construct a concept map linking the progression from Data → Data Object → Data Structure → Abstract Data Type (ADT) → Implementation in C. Identify relevant sub-concepts such as structures, pointers, and operations, and establish meaningful relationships among them. Organize the concept map hierarchically and justify how each level contributes to efficient data organization, processing, and improved system performance.



- **Data** represents raw patient information such as temperature readings, medical history, and diagnostic details.
- A **Data Object** represents a meaningful individual piece or record of patient information.
- **Data Structures** organize these data objects for efficient storage and processing.
- **Structures** can group different types of patient information into a single record.
- **Pointers** store memory addresses and support efficient access to data.
- An **Abstract Data Type (ADT)** defines the data and operations that can be performed on it.
- Common ADT operations include **insertion, deletion, searching, and updating**.
- The ADT can be implemented in **C using structures, pointers, arrays, and functions**.
- This hierarchical organization creates clear and meaningful relationships between each level.
- Therefore, suitable data structures and ADTs improve **data organization, processing efficiency, and overall system performance**.

Q2 A smart city traffic management system handles data related to vehicle movement, road networks, and signal control. Construct a concept map connecting Primitive Data Structures → Non-Primitive Data Structures → Linear → Non-Linear Structures. Include examples such as arrays, linked lists, trees, and graphs, and illustrate their applications in traffic management. Explain the relationships between these structures and justify their classification based on functionality and efficiency.



- **Primitive data structures** are basic data types used to store simple traffic information.
- Examples include **int for vehicle count, float for vehicle speed, and char for signal status.**
- These basic types are used to construct **non-primitive data structures.**
- Non-primitive data structures are classified into **linear and non-linear structures.**
- **Linear structures** arrange data elements sequentially.
- **Arrays** can store vehicle data or traffic counts, while **linked lists** can manage changing vehicle records.
- A **queue** can represent vehicles waiting at a traffic signal.
- **Non-linear structures** represent hierarchical or network-based relationships.
- **Trees** can represent road or traffic hierarchies, while **graphs** can represent road networks and connections between junctions.
- Therefore, this classification helps select suitable structures for **vehicle management, signal control, route planning, efficient searching, and traffic processing.**